

Set notation and sample space



- 1 List all the possible outcomes when a coin is flipped.
- 2 A standard six-sided die is rolled.
 - a List the sample space.
 - b List the sample space for rolling a number strictly less than 3.
 - c List the sample space for rolling a number divisible by 3.
 - d List the sample space for rolling an even number.

- 3 Uther rolled a standard six-sided die.
 - a List all the numbers that the die may land on.
 - b Uther rolled a number less than 3. List all the numbers that he could have rolled.



- 4 List all the elements in the following sets:
 - a $\{x : x \text{ is a positive even integers that are less than } 16\}$
 - b $\{x : x \text{ is an integer between } -8 \text{ and } -3 \text{ (not inclusive)}\}$
 - c $\{x : x \text{ is an odd whole number less than } 13\}$
 - d $\left\{1, \frac{1}{10}, \frac{1}{100}, \dots, \frac{1}{100\,000}\right\}$
 - e $\{2, 4, 8, \dots, 64\}$

- 5 Consider the following sets:

$$A = \{1, 2, 4, 8, 16\}, \quad B = \{1, 2, 4, 8\}$$

List all the elements in:

- a A or B
- b Both A and B

Set notation (Extended)

- 6 Consider the set $A = \{2, 4, 6, 8\}$. Construct set builder notation for A .

7 Consider the following sets:



$$A = \{8, 12, 16, 20, 24, 28\}, B = \{8, 12, 16, 28\}, C = \{12, 16\}, \text{ and } D = \{4, 8, 12, 16, 20, 24, 28\}$$

Determine whether the following statements are true or false:

- a $B \subset D$ b $C \subset B$ c $A \subset B$ d $\emptyset \subset D$

8 Write $\subseteq$ or $\not\subseteq$ to make each of the following statements true:

- a $\{1, 2, 4\} \subseteq \{1, 2, 4, 7, 8\}$ b $\{8, 9\} \subseteq \{3, 4, 7, 8, 9\}$
c $\{3, 4, 9\} \subseteq \{2, 4, 7, 8, 9\}$ d $\{1, 3, 8\} \subseteq \{1, 4, 7, 8\}$

9 Write $\in$ or $\notin$ to make each of the following statements true:

- a $5 \in \{2, 5, 6, 9\}$ b $11 \in \{4, 8, 12, 14\}$

10 Consider the following sets:

- $A = \{\text{letters of the word 'parallelogram'}\}$
- $B = \{\text{letters of the word 'gallop'}\}$
- $C = \{\text{letters of the word 'llama'}\}$

- a State the set of elements contained in both A and B .
b Is B a proper subset of A ?
c State the set of elements contained in A or C .

11 Explain the meaning of "the complement of set A ."

12 A set is defined as $A = \{x : 0 \leq x \leq 5 \text{ and } x \text{ is an integer}\}$.

- a List the elements of set A . b Is set $B = \{2, 3, 5\}$ a subset of set A ?

13 Consider the sets $A = \{x : -6 < x < 6 \text{ and } x \text{ is an integer}\}$ and $B = \{x : -1 \leq x \leq 10 \text{ and } x \text{ is an integer}\}$.

- a Find $n(A)$. b Find $n(B)$. c Find $n(A \cap B)$. d Find $n(A \cup B)$.

14 State whether the following points are in the set $A = \{(x, y) : y = 5x - 7\}$:

- a $(9, 40)$ b $(2, 3)$ c $(0, 7)$ d $\left(\frac{3}{2}, \frac{1}{2}\right)$

15 Consider the sets $A = \{(x, y) : y = 3x - 4\}$  $B = \{(x, y) : y = 2x + 6\}$.

a Explain why $n(A \cap B) = 1$.

b Find $A \cap B$.

Operations on sets

16 Explain the meaning of the following statements:

a Union of set A and set B .

b Intersection of set A and set B .

17 P and Q are sets of vegetable types:

$P = \{\text{carrots, cauliflowers, beans}\}$; $Q = \{\text{cauliflowers, potatoes}\}$

There are no other vegetable types in universal set U .

a Is $P \cup Q$ the set of all vegetable types?

b List the elements in the set $P \cap Q$.

18 Suppose set $A = \{3, 4, 5, 6, 7\}$ and set $B = \{3, 7, 8, 9\}$. Find $A \cap B$.

19 List the elements of $A \cap B$ given the following sets:

a $A = \{\text{even numbers}\}$ and $B = \{\text{square numbers less than 100}\}$.

b $A = \{\text{multiples of 5}\}$ and $B = \{\text{positive numbers less than 50}\}$

20 If A is the set of factors of 24, and B is the set of factors of 36, then list the elements of:

a $B \cup A$

b $A \cap B$

21 Set A is the set of possible outcomes from rolling a standard die, and set B is the set of possible outcomes from rolling an eight-sided die. List the elements of the following sets:

a A

b B

c $A \cap B$

d $A \cup B$



Complement of sets (Extended)

22 P and Q are sets of flower varieties:

$$P = \{\text{roses, lillies, daisies}\}; Q = \{\text{lillies, sunflowers}\}$$

There are no other flower varieties in universal set U .

a List the elements in the following sets:

i Q'

ii $Q \cap P$

iii $Q \cup P$

iv $Q' \cap P$

b Is $(P \cup Q)'$ empty?

23 The sets $V = \{21, 8, 30, 9, 28\}$ and $W = \{21, 8, 30, 9, 28, 7, 13, 12, 26\}$ are such that there are no other elements outside of these two sets.

a Is V a proper subset of W ?

b State $n(V)$, the order of V .

c List the elements of V' .

d List the elements of the universal set.

e Find W' .

24 Consider the following sets:

$$A = \{1, 2, 3, 4, 5, 6, 7\}, B = \{1, 2, 3, 4\}$$

If there are no elements contained outside of these sets, find:

a B'

b A'

25 Consider the following sets:

$$A = \{\text{people who like football}\}$$

$$C = \{\text{people who like swimming}\}$$

$$B = \{\text{people who like softball}\}$$

$$D = \{\text{people who do not like any of these}\}$$

a Describe set B' .

b Describe set D' .

26 Consider the sets $A = \{x : -7 \leq x \leq 10 \text{ and } x \text{ is an integer}\}$, $B = \{-2, 4, 8\}$ and $C = \{3, -5, 11\}$.

a Is set B a subset of set A ?

b Is set C a subset of set A ?

c List the elements in the set $A \cap C$.

d List the elements in the set $A' \cap C$.

e Find $n(A \cup C)$.

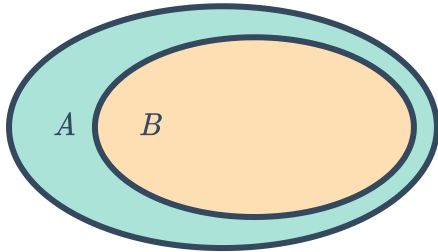
f Find $n(A \cap C')$.



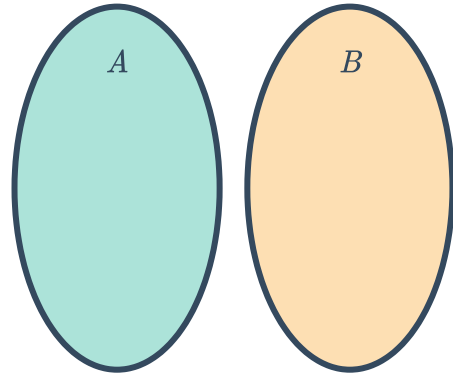
Venn diagrams

27 For each of the following, identify the Venn diagram that best represents the sets A and B :

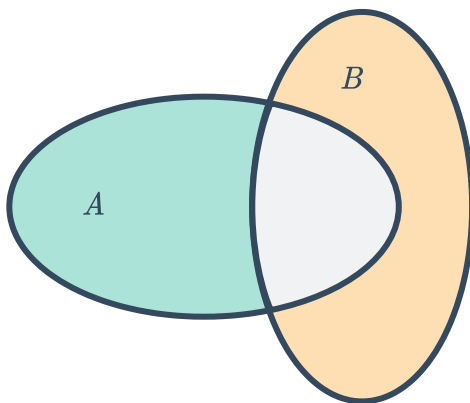
Venn diagram 1



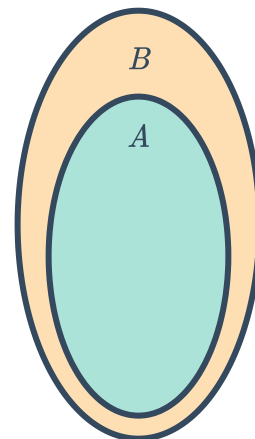
Venn diagram 2



Venn diagram 3



Venn diagram 4



- a** $A = \{\text{Earth}\}$, $B = \{\text{Planets}\}$
- b** $A = \{6, 8, 4, 2, 7\}$, $B = \{3, 6, 1\}$
- c** $A = \{\text{w, i, n, d}\}$, $B = \{\text{e, a, r, t, h}\}$
- d** $A = \{\text{Animals found in Australia}\}$, $B = \{\text{Animals found in NSW}\}$

28 We are interested in the colour of a card randomly drawn from a standard deck. Draw a Venn diagram to illustrate this.

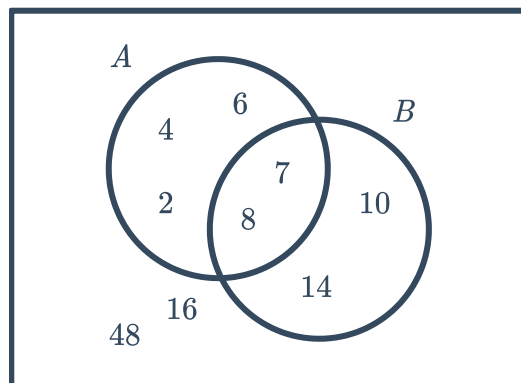
29 Consider the Venn diagram:



List the elements in:

a $A \cap B$

b $A \cup B$



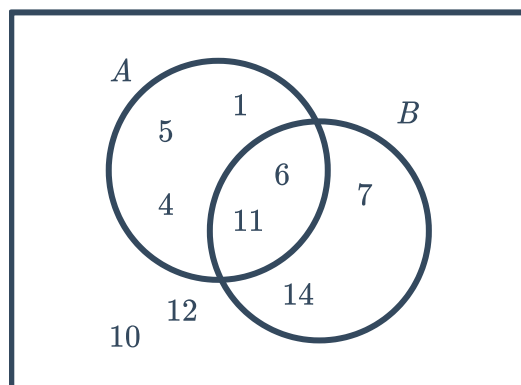
Venn diagrams (Extended)

30 Consider the Venn diagram:

List the elements in:

a A'

b B'



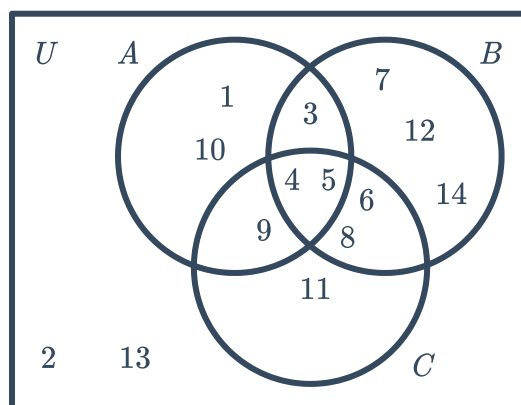
31 Consider the Venn diagram:

List the elements in:

a A

b U

c B'

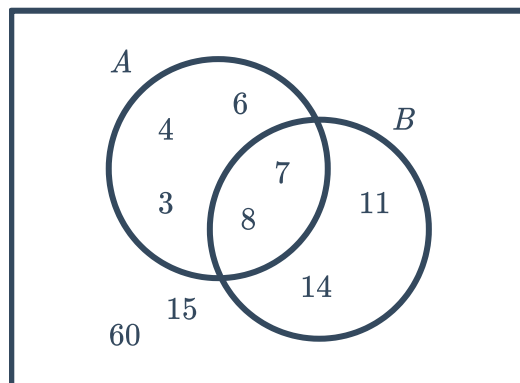


32 Consider the Venn diagram:



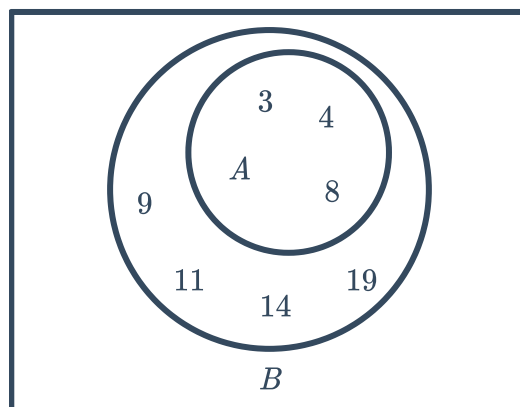
List the elements in:

- a $A \cap B'$ b $(A \cup B)'$



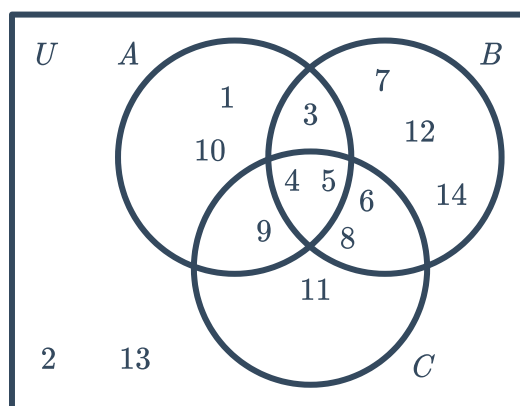
33 Consider the following Venn diagram:

- a Is A a subset of B ?
b Is A a proper subset of B ?



34 Consider the following diagram:

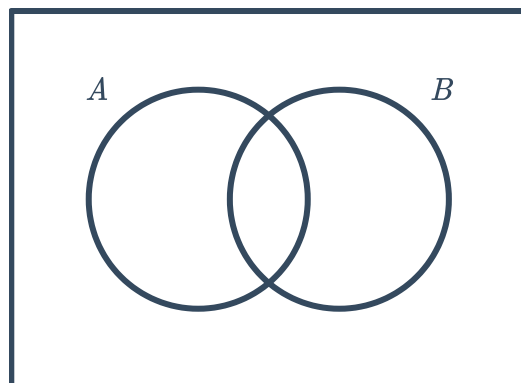
- a Find the set $A \cap C$.
b Find the set $(B \cap C)'$.
c Find the set $A \cap B \cap C$.
d Find the set $A \cap (B \cup C)$.
e Find the set $(A \cap B)'$.



35 Consider the following Venn diagram:



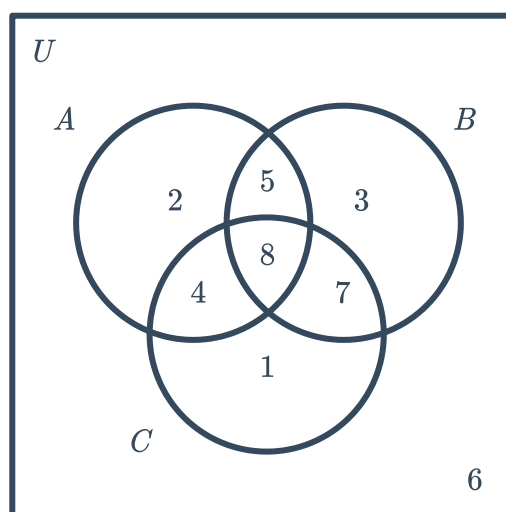
- a Is $(A \cap B)'$ equal to $A' \cup B'$ for all sets?
- b Is $A' \cap B'$ equal to $A \cup B'$ for all sets?



36 Consider the following Venn diagram:

Find the elements in the following:

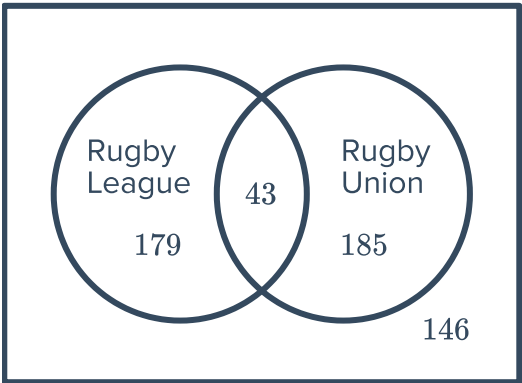
- a $A \cap B' \cap C'$
- b $A \cap B \cap C'$
- c $A' \cap B \cap C'$
- d $A \cap B' \cap C$
- e $A \cap B \cap C$
- f $A' \cap B \cap C$
- g $A' \cap B' \cap C$
- h $A' \cap B' \cap C'$



37 The Venn diagram shows the number of students in a school playing Rugby League (L), Rugby Union (U), both or neither.

a Complete the table of values below:

	Play Rugby League	Don't play Rugby League
Play Rugby Union		
Don't play Rugby Union		



b Find the following:

- i $n(L \cap U')$
- ii $n(L)$
- iii $n(U)$
- iv $n(U \cap L')$
- v $n(L')$
- vi $n(U')$